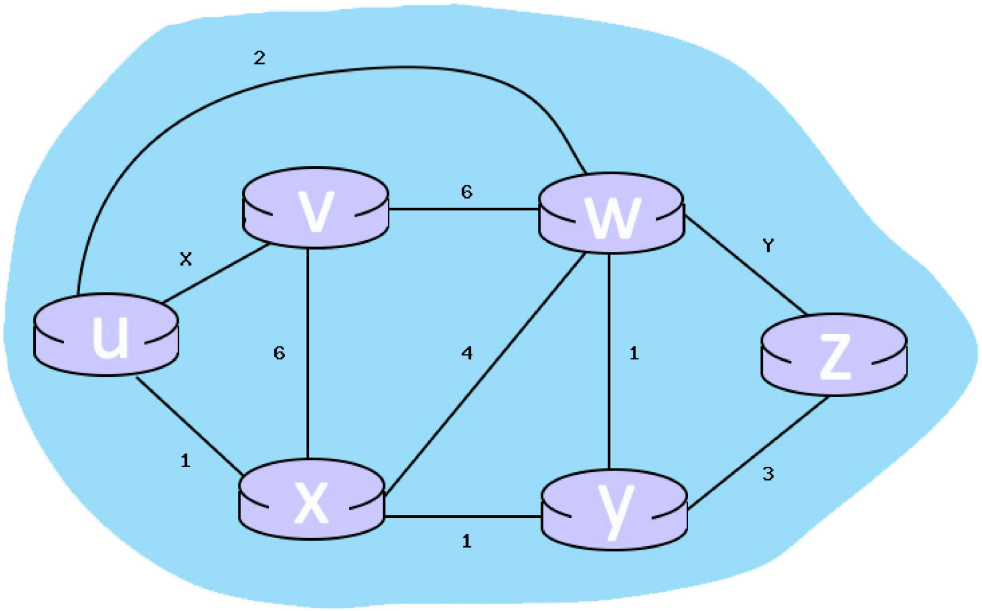
INTERACTIVE END-OF-CHAPTER EXERCISES

DIJKSTRA'S LINK STATE ALGORITHM - ADVANCED

Consider the incomplete 6-node network shown below, with the given link costs.



Consider the completed table below, which calculates the shortest distance to all nodes from X:

Node	Shortest distance from X	Previous Node
X	0	n/a
U	1	X
Y	1	X
V	2	U
W	2	Y
Z	4	Y

QUESTION LIST

1. For link X, what is the cost associated with this link? If the answer can't be determined given the information, respond with 'n/a'

2. For link Y, what is the cost associated with this link? If the answer can't be determined given the information, respond with 'n/a'

SOLUTION

1. The prior node in the path to V is U, and we know the shortest distance of both V (2) and U (1), so 2 - 1 = 1 which is X.

2. Because the link is never used, we cannot determine the value of Y, so the answer is n/a.

QUESTION 1 OF 2

For link X, what is the cost associated with this link? If the answer can't be determined given the information, respond with 'n/a'

Answer

CHECK HINT SKIP

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